



CSE499A
Section 03

CSE499A PROJECT REPORT

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Table of Contents

Abstract

Keywords

1. Introduction
2. Current KYC Landscape in Bangladesh
3. Literature Review
 - 3.1 Relevant Research Papers
4. Analysis and Comparison across Bangladeshi Bank Policies
5. Case Studies Around the Globe
 - 5.1 Case Study 1: India's Aadhaar-Based e-KYC System
 - 5.2 Case Study 2: EU eIDAS Regulations for Digital Identity
 - 5.3 Case Study 3: Singapore's MyInfo – "Tell Us Once"
6. Methodology
 - 6.1 Approach
 - 6.1.1 Core Design Philosophy
 - 6.2 Architectural Framework
7. Technology Used
 - 7.1 For Blockchain Infrastructure
 - 7.2 For Development
 - 7.3 Cryptographic Components
 - 7.4 Data Management
8. Implementation
 - 8.1 Core Data Structures
 - 8.2 Credential Lifecycle Management
 - 8.3 Verification Process
 - 8.5 Revocation Mechanism
9. Privacy and Security Implementation
 - 9.1 Setup and Execution
10. Scope of Work in CSE499B
11. Challenges
12. Conclusion
13. Reference

"A Blockchain-Based Uniform e-KYC System for Banks in Bangladesh"

ABSTRACT:

Banking is a formidable force in driving the economy throughout the world, even in Bangladesh. Customer onboarding is a crucial step in a bank's service lifecycle, conducted through Know Your Customer (KYC) policies. Although Bangladesh Bank provides a guideline for KYC, banks often add their own requirements. As a result, whenever a customer wants to open an account at a different bank, they must go through the full KYC process again. This repeated verification is time-consuming for both customers and banks, discouraging people from opening accounts at multiple institutions. Our proposed solution addresses these problems through an e-KYC platform built on smart contracts. Once verified at one bank, a customer does not need to be re-verified elsewhere, as the smart contract securely holds their records and onboarding history. With proper consent, banks can access the relevant information, ensuring privacy. Smart contracts are also tamper-proof, providing proof of verification and safeguarding against fraud. Overall, the system ensures secure, transparent, and efficient onboarding.

Keywords: Blockchain, Smart Contracts, e-KYC, Bangladesh Banking System, Customer Onboarding, Data Privacy, Anti-Money Laundering (AML), Combating Financing of Terrorism (CFT), Bangladesh Financial

Intelligence Unit (BFIU), Regulatory Compliance, Digital Identity Management, Self-Sovereign Identity (SSI), Decentralized Identity.

1. Introduction:

Know your Customer(KYC) is a mandatory process for banks and other financial institutions to verify the identity of clients. It is an essential part of a bank's Anti-Money Laundering(AML) act and also Combating the financing of Terrorism(CFT). The main purpose of KYC lies in safeguarding the bank from getting involved in terrorism financing, fraudulent deals and money laundering. Seeing how important KYC is to our banks, it is important that the KYC system, regardless of which bank the KYC is being dispersed from, has to be rigorous in sorting out the good clients from the bad. But, it is a matter of sorrow that most banks do not follow a uniform KYC system. Yes the baseline of the kYC process is adhering to the policies set by Bangladesh Bank and these KYC processes also depend on the Money Laundering Prevention Act, 2012, and the Anti-Terrorism Act, 2009. The BFIU is responsible for issuing guidelines and monitoring compliance . Despite these there are additions to these policies which vary bank by bank(mainly on the KYC principles). As already stated in our abstract, the manual KYC process is cumbersome, lengthy in duration and not secured enough. On top of that there are also risks of fraud and duplication in manual forms of KYC systems. And the systems that are digital are not used by the local banks at all. So, at the end of the day most banks are using a manual form of KYC process. Considering all the stated factors and problems we proposed a solution or a starting point of a

solution. Our proposed smart contract based e-KYC will combat all of the stated problems and on top of that make our KYC system more efficient. Along with that, the problem of different banks having different thresholds for daily transactions won't be an issue since the transactions will take place from the bank platform, not our smart contract platform. That is because our smart contract platform will just deal with the verification and onboarding step of the customer, not the transactions of money. In this way, the banks do not have to worry about setting their own daily threshold. In a nutshell, how we will accomplish these tasks in building the platform along with the policies that adhere to the laws and regulations of Bangladesh will be explored in this report.

2. CURRENT KYC LANDSCAPE IN BANGLADESH:

In Bangladesh KYC is a vital process in onboarding a customer in terms of seeing if the customer is legit, non-fraudulent and adhering to the policies of CFT and AML. Although the KYC process is crucial for the banks, they are still mostly relying on the manual KYC processes. These manual KYC processes require several business days to check the validity of customers and the banks who have their digital KYC system do not even touch it. Despite most banks not following the modern standard of KYC processes Pubali Bank, Mercantile Bank and MFS like bKash, Nagad did adopt e-KYC for onboarding their customers. For instance, Pubali Bank uses their PI app which has an instant account opening feature. Mercantile bank has MBL rainbow for remote account opening with their NID card. Despite these being good examples, the problems of privacy ownership and having a

system where the users do not have to open accounts every time they want to transact from a different bank persists.

3. LITERATURE REVIEW:

3.1 RELEVANT RESEARCH PAPERS:

In the paper "***Enabling Trust and Privacy Preserving e-KYC System Using Blockchain***" authors Mrs. M. Ashwitha Reddy, R. Naveen, Potlapalli Neeraj Rao, K. Sree Kalyani proposes a blockchain based e-KYC system called "e-KYC Trustblock" to address security and privacy issues in cloud-based e-KYC processes used by banks and financial institutions. Around the world traditional eKYC relies on cloud storage, leading to risks like single points of failure, limited traceability and unauthorized access by cloud providers. The authors integrate blockchain with Cipher-Policy Attribute-Based Encryption to enforce user consent, encrypt sensitive data, and enable fine-grained access control. Key features include: user registration and upload of KYC documents to InterPlanetary File System(IPFS) via **smart contracts**, admin approval of document validity, symmetric encryption for client files combined with public-key encryption and policy updates for re-encryption.

Then in the paper, "***A Blockchain-Based Self-Sovereign Identity System for KYC Processes***" authors Chengzu Dong, Aiting Yao, Zhiyu

Xu, Meiqu Lu, Frank Jiang, Shiping Chen, Xiao Liu introduced a blockchain based eKYC system using Self-Sovereign Identity to enhance privacy, reduce costs and ensure compliance with Anti-Money Laundering and Terrorism Financing regulations. Traditional KYC is complex, costly and prone to non-compliance penalties. The system introduced by the authors also leverages SSI roles(holder, verifier and issuer) which is according to **Wide Web Consortium standards**, allowing users to control their KYC data. In the paper, the authors explored immutable consent recording on blockchain, real time regulatory updates, proactive monitoring of KYC data expiration predominantly. We found this paper to be the most inspirational for our system .

We also explored the paper “***Know Your Customer (KYC) Implementation with Smart Contracts on a Privacy-Oriented Decentralized Architecture***” authored by Nikolaos Kapsoulis, Alexandros Psychas, Georgios Palaiochrassas, Achilleas Marinakis, Antonios Litke, Theodora Varvarigou. The authors explained and proposed a decentralised KYC system using smart contracts on a permissioned blockchain(Quorum) integrated with IPFS for privacy preserving identity management. Blockchain enables transparent, low cost micropayments and content protection but requires the step of robust KYC to link cryptographic identities to real world ones.

We also reviewed the paper provided by our supervisor Dr. Rashedur Rahman which was titled “A Systematic Literature Review of Blockchain-Based e-KYC Systems

” authored by Md. Abdul Hannan, Md. Atik Shahriar, Md. Sadek Ferdous, Mohammad Javed Morshed Chowdhury, Mohammad Shahriar Rahman. Prisma model was used to analyse existing research of blockchain and eKYC. It was mentioned that blockchain’s **immutability** aids in **transparency** and reduces redundancy.

Moving on we went through the paper titled “Hassle-Free and Secure e-KYC System Using Distributed Ledger Technology” authored by Bharti Pralhad Rankhambe and Dr. Mrs. Harmeet Kaur Khanuja from the Department of Computer Engineering at MMCOE, Pune, India. This paper deep dived into the integration of blockchain technology into the banking industry's KYC processes. The proposed system by the author leverages the use of blockchain to store and track customer records along with eliminating middlemen, reducing errors, and enabling one-time verification reusable across multiple financial institutions. The fact that the **verification is one time** across multiple institutes is what caught our attention since this aligns with our goal of implementing our eKYC platform.

To further our idea on DIDs and SSI we read the paper titled “Blockchain-Enabled Decentralized Identity Management: The Case of Self-Sovereign Identity in Public Transportation” authored by Lukas Stockburger, Georgios Kokosioulis, Alivelu Mukkamala, Raghava Rao Mukkamala, and Michel Avital from Copenhagen Business School and Kristiania University College. Although their work was on the public transportation sector; the ideas surrounding identity management

fundamentals, distinguishing digital identities from psychological concepts and highlighting risks in centralized systems was vital to our thought process regarding our system. The authors proposed SSI—enabled by blockchain—as a user-controlled alternative, where individuals manage credentials via wallets without third-party reliance. The author’s low-fidelity prototype demonstrates how users can use standardized credentials (e.g., travel passes) across European networks, eliminating multiple cards and enhancing data control. Throughout the paper it was evident that Blockchain ensures trust among stakeholders (e.g., transport authorities) via transparency and immutability, while SSI principles (e.g., data minimization) protect privacy.

In the actual blockchain based system that we started building(explained later on), we used **keccak256** hashing algorithm instead of SHA256 simply because it is more suited to the blockchain system that we are building. So, we also reviewed the paper titled “A Pseudorandom Number Generator with KECCAK Hash Function” authored by A. Gholipour, S. Mirzakuchaki. The paper mentioned above introduces a pseudorandom number generator algorithm based on the KECCAK hash function, a SHA-3 competition candidate known for its sponge construction. The algorithm uses a random seed to produce bit sequences by iteratively applying **KECCAK-f[1600]**. This processes inputs through absorbing and squeezing phases with permutation steps(θ , ρ , π , χ , ι). The generated sequences are tested for randomness using **NIST statistical Test Suite**. This passes the 15 tests(such as frequency, runs, FFT) with high success ratios and P-values indicating strong pseudo-randomness. The approach addresses older hash-based PRNGs like those using MD5 or SHA-1. And simply for this

particular case we decided to use KECCAK 256 which also happened to be built in the solidity language framework.

Initially when we started reading our research papers for our ideation and getting inspiration we came across “A New Approach to Client Onboarding Using Self-Sovereign Identity and Distributed Ledger” which was authored by Reza Soltani, Uyen Trang Nguyen and Aijun An from the Department of Computer Science and Engineering at York University, Toronto, Canada. In the paper it was mentioned how client onboarding and Know Your Customer (KYC) processes are slow and expensive. The fact that modern identity management models use centralised systems in handling user data which is insecure. Such tendencies lead to breaches like Equifax. And to avoid such disasters a decentralised system was proposed which will not keep user data centrally . Addition of GDPR compliance and cost cut of KYC build up is another motivational aspect in adopting a decentralised system because this will reduce redundancy, increase efficiency and also ensure privacy. With all these in mind, the authors proposed the KYC2 framework which is a digital onboarding framework using Hyperledger Indy. The reason **Hyperledger Indy** was used for their work is because it complies with the SSI’s ten principles(which includes control from a round robin voting algorithm called byzantine protocol and transparency because the view is open for all). Also with that DID(decentralised identification), Verifiable Credentials were easy to disburse and top of that Hyperledger Indy was not proprietary backed. Moreover, the paper also gave reviews on decentralised PKI efforts and SSI projects such as Blockcert, ShoCard, OpenBadges etc .We would have picked this paper as our Bible if only

Hyperledger was still supported, which is not the case now. But we took the essential lessons from this paper and implemented them in our project.

4. Analysis and Comparison across Bangladeshi Bank Policies:

We conducted exploration and comparison of KYC documentation requirements from various Bangladeshi banks Anti Money Laundering (AML) and Combating Financing of Terrorism (CFT) policies.

We started our analysis with an inquiry into the latest KYC details from the **First Security Islami Bank PLC (FSIBL)** policy document [AML-CFT-Policy-2025.pdf], . And later more analysis with three additional banks policies, **Agrani Bank** [AML_CFT_FINAL_VERSION.pdf], **Bank Asia**[Money-Laundering_Terrorist_Financing_Risk_Management_Guideline-2025.pdf], and **Sonali Bank** [AML Guidelines_3.0.pdf]. The analysis we conducted aligned with Bangladesh Financial Intelligence Unit (BFIU) guidelines under the Money Laundering Prevention Act 2012 (amended 2015).

In the FSIBL document, Chapter 7 (Customer Due Diligence) carries the core area detailing KYC procedures within the bank's CDD framework, including customer identification, verification, risk based approaches and ongoing monitoring. Additionally we checked Annexure B, C and D which reflected the bank's policy, incorporating BFIU requirements, the Money Laundering Prevention Act 2012 and other related rules.

The comparison we conducted on FSIBL with other banks (Agrani Bank, Bank Asia and Sonali Bank), we highlighted some similarities, such as reliance on government issued IDs, income proofs, and utility bills, all under a risk based framework with Enhanced Due Diligence (EDD) for high risk customers like influencers or any sort of politicians.

The difference included Bank Asia's emphasis on e-KYC (beginning around mid 2021) with biometric integration, while others focused more on the traditional paper work method. We also found variations in document specifics like Bank Asia adding e-TIn and update frequencies (1-5 years based on risk). Rest of everything is very much similar ensuring alignment with BFIU standards.

BFIU's Electronic Know Your Customer (e-KYC) guidelines from Circular 25, which emphasize a risk based approach with digital tools like biometric verification, NID integration and video based onboarding. Key requirements included customer identification through NID, passport, source of funds verification (salary certificates, trade licenses etc.), and address proof (utility bills etc.) for individuals and joint accounts. EDD (Enhanced Due Diligence) for high risk categories like Politically Exposed Persons (PEPs) and ongoing monitoring, with records retained for at least 5 years.

5. CASE STUDIES AROUND THE GLOBE:

5.1 Case Study 1: India's Aadhaar-Based e-KYC System

India's **Aadhaar-based e-KYC** is part of the country's national digital identity infrastructure managed by **UIDAI**, designed to simplify access to services by providing instant, paperless verification. Before Aadhaar, KYC

processes were manual, document-heavy, costly, and exclusionary, leaving millions of low-income citizens without access to banking and welfare benefits. The system works by linking an individual's **12-digit Aadhaar number** with **biometric or OTP-based authentication**, which is verified against a central database; once authenticated, a digitally signed KYC token is issued to service providers as valid proof of identity. Its **positive impacts** include reducing the cost of verification from about ₹1000 to ₹5–15, cutting onboarding time from weeks to minutes, enabling financial inclusion through schemes like PMJDY, and saving the government billions by direct benefit transfers to Aadhaar-linked accounts. However, it faces serious **challenges** such as privacy concerns over centralized biometric storage, data security breaches from leaks and unauthorized access, and exclusion issues where authentication failures prevent vulnerable populations from accessing essential services. These tensions have led to legal scrutiny; notably the **2018 Supreme Court judgment** that restricted its mandatory use while upholding its role in welfare, highlighting the delicate balance between efficiency, inclusion, and rights protection.

5.2 Case Study 2: EU eIDAS Regulations for Digital Identity

The **EU's eIDAS Regulation (2014)** established a uniform legal and technical framework to address Europe's previously fragmented digital identity landscape, where national eIDs lacked cross-border recognition and electronic signatures carried uneven legal weight. It introduced two pillars: **electronic identification (eID)**, enabling mutual recognition of national eID schemes once 'notified' to the European Commission and

trust services, which legally equate e-signatures, seals, timestamps, and certificates with their paper counterparts. The system operates through interoperability nodes, allowing a citizen of one EU state to authenticate with their home eID system and securely access services in another, without sharing sensitive credentials directly. This has delivered **positive impacts**, including cross-border access to education, business, and public services, legal certainty for digital transactions, boosted e-commerce and banking, and strengthened user control and privacy through data minimization principles. Yet, **adoption challenges** emerged due to slow rollout, limited private sector uptake, and technical integration hurdles compared to simpler commercial logins. To address this, **eIDAS 2.0** introduces the **European Digital Identity Wallet (EUDI Wallet)**, which will let citizens store their eID alongside digital documents (licenses, diplomas, prescriptions), provide selective disclosure, and enable qualified e-signatures for both public and private services, thereby reinforcing trust, privacy, and usability in Europe's digital single market.

5.3 Case Study 3: Singapore's MyInfo – “Tell Us Once”

Singapore's **MyInfo**, launched in 2016 by **GovTech** and the Ministry of Finance, was developed to overcome the inefficiencies of citizens repeatedly re-entering personal details across digital platforms, which led to errors, delays, and higher onboarding costs. Built on the National Digital Identity framework and integrated with **Singpass authentication**, MyInfo acts as a digital vault of government-verified data (e.g., name, NRIC, address, employment, tax records). The process is consent-based: citizens

log in via Singpass, review which data fields are requested, and authorize their release to agencies or businesses, ensuring transparency and control. Its **positive impacts** include drastically reducing form-filling time, improving accuracy by sourcing data from authoritative systems, lowering the risks of document misuse, and enabling banks, insurers, and telcos to conduct faster, cheaper KYC onboarding. Adoption has grown rapidly, now supporting over 1,600 services and millions of monthly transactions, reflecting its success as a public-private digital infrastructure. However, **challenges** remain including cybersecurity risks associated with centralized storage, concerns over function creep and surveillance, and the need to ensure digital inclusion for all citizens. To strengthen resilience, recent upgrades such as **MyInfo v5**, **FAPI 2.0 standards**, and upcoming features like digital signing, QR-based verification, and audit trails position MyInfo as an evolving ecosystem that balances efficiency, security, and user trust.

6. METHODOLOGY:

6.1 Approach:

What we envisioned to our blockchain eKYC platform to be, emphasises on the fact that we will hold a decentralised management approach which will leverage smart contracts to create a trustworthy, privacy-preserving identity verification system which will be efficient in eliminating the problems these manual KYC processes hold in our country. At first we were heavily

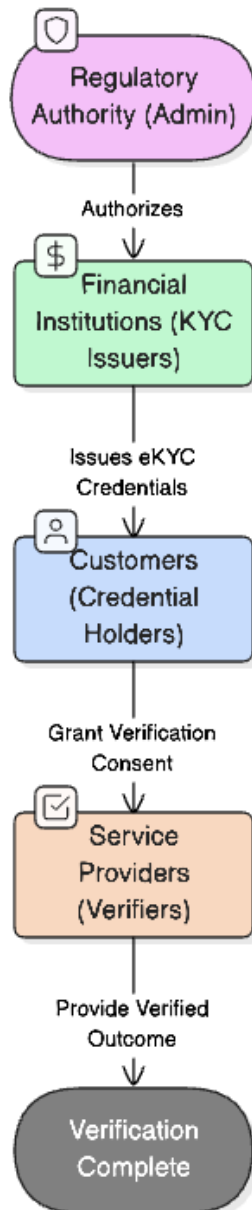
considering using Hyperledger Indy to implement our blockchain system but in the end we could not go for it simply because that project was not supported anymore in 2025. And all of its subsidiaries also lost support. So we initially thought of using either a Smart Contracts based approach which is based on Ethereum or Solana . But we decided to go with the Ethereum based smart contracts approach simply because that is well supported and is also proof of stake. Meaning, the banks with the most stake will have a leverage over the other banks in terms of priority. This is better than proof of work because in POW the banks would rather try to do as many transactions as possible on the blockchain network rather than actually having a share on the blockchain network themselves . So we avoid proof of work based blockchain technologies and chose Ethereum as our base.

6.1.1 Core Design Philosophy:

Our platform implements a hash based verification system where sensitive customer data remains off-chain(much like oracles in blockchain networks) and at the same time cryptographic proofs are stored in the blockchain. This approach ensures data privacy while maintaining the immutable audit trail required for regulatory compliance.

6.2 Architectural Framework:

The hierarchical structure given below ensures proper governance while at the same time maintaining decentralized verification capabilities, addressing the need for both regulatory oversight and user privacy control.



For better understanding, regulatory authority is the admin. Admin is the one who will deploy the smart contract network to begin with. For our system, Bangladesh Bank will be the admin. The issuers are the Banks that are licensed under Bangladesh Bank. So, issuers will be accounted for when a user or customer is verified from that particular bank's end. The Verifiers are the banks that will verify an already verified user(done from a certain bank at the beginning). The customers are termed as Holders in the system since they will be holding their Credentials.

7. TECHNOLOGY USED:

7.1 For Blockchain Infrastructure:

Ethereum Blockchain: Selected as the underlying distributed ledger for its mature smart contract ecosystem and widespread adoption

Solidity Programming Language: Version ^0.8.0 used by us for smart contract development which provided built-in overflow protection and enhanced security features

EVM : Ethereum Virtual Machine Provides the runtime environment for smart contract execution

7.2 For Development:

Remix IDE: Browser-based integrated development environment for smart contract development, compilation, and deployment

OpenZeppelin Libraries: Industry-standard smart contract libraries for secure implementation patterns

7.3 Cryptographic Components:

Keccak-256 Hashing: Used for generating unique credential identifiers and storing document hashes

Elliptic Curve Digital Signatures: Ethereum's native signature scheme for transaction authentication

Merkle Tree Structures: For efficient batch verification of multiple credentials

7.4 Data Management:

On-chain Storage: Limited to credential hashes, metadata, and relationship mappings

Off-chain Storage: IPFS or traditional encrypted databases for actual KYC documents

Event Logging: Ethereum events for creating immutable audit trails

8. IMPLEMENTATION:

8.1 Core Data Structures:

Our platform utilised a structured approach to credential management:

```
struct Credential {  
    bytes32 credentialHash;  
    address issuer;  
    address holder;  
    uint256 issuedAt;  
    bool isRevoked;  
    uint256 credentialType; // 1 = Student ID, 2 = KYC  
}
```

Our platform used a role-based access control through solidity modifiers:

Administrative Control: `onlyAdmin` modifier restricts critical functions to regulatory authorities

Issuer Authorization: `onlyAuthorizedIssuer` ensures only licensed financial institutions can issue credentials

Holder Consent: Built-in consent mechanisms protect customer privacy rights

```

modifier onlyAdmin() {
    require(msg.sender == admin, "Only admin");
    _;
}

modifier onlyAuthorizedIssuer() {
    require(authorizedIssuers[msg.sender], "Not authorized issuer");
    _;
}

```

The functions that will use these modifiers will be only bounded as such. So if someone is admin only they can access or use that particular function. Same case for the onlyAuthorisedIssuer.

8.2 Credential Lifecycle Management:

Admin Addition:

The admin is declared by default as they deploy the smart contract. When the smart contract is deployed simply the **constructor** of the smart contract code is called and the admin is set.

```

constructor() {
    admin = msg.sender;
}

```

Bank adding procedure:

Only an admin, which is the case when an entity deploys the smart contract, can add registered banks which are issuers or verifiers in our case. The admin is Bangladesh Bank in our perceived simulation.

```
function addAuthorizedIssuer(address _issuer) public onlyAdmin {  
    authorizedIssuers[_issuer] = true;  
}
```

Here, as per the modifier named **onlyAdmin** can be the one adding an issuer to the authorizedIssuers map.

Issuing Process:

Authorized issuer collects and validates KYC documents off-chain. The document data is hashed using **KECCAK256** algorithm

```

function issueCredential(  infinite gas
    bytes32 _credentialHash,
    address _holder,
    uint256 _credentialType
) public onlyAuthorizedIssuer returns (bytes32) {
    bytes32 credentialId = keccak256(
        abi.encodePacked(_credentialHash, _holder, block.timestamp)
    );

    credentials[credentialId] = Credential({
        credentialHash: _credentialHash,
        issuer: msg.sender,
        holder: _holder,
        issuedAt: block.timestamp,
        isRevoked: false,
        credentialType: _credentialType
    });

    holderCredentials[_holder].push(credentialId);

    emit CredentialIssued(credentialId, msg.sender, _holder, _credentialType);

    return credentialId;
}

```

As we can see the function has a modifier which is **onlyAuthorizedIssuer**. So an authorised issuer can only invoke this function. And in our system authorized issuers will be the registered banks. The smart contract as per the function snippet given above generates a unique credential ID using KECCAK256. Here credential metadata stored on-chain with hash reference. Afterwards, as we can see Credential ID added to holder's credential registry here holderCredentials is like a map which we defined in code. **CredentialIssued** event is then emitted for audit trail as we can see at the end of the function.

8.3 Verification Process:

Here, the service provider requests KYC verification from the customer.

The customer grants explicit consent through `grantConsent()` function.

```
function grantConsent(address _verifier, bytes32 _credentialId) public { 4813 gas
    require(credentials[_credentialId].holder == msg.sender, "Not credential holder");
    emit ConsentGranted(msg.sender, _verifier, _credentialId);
}
```

At the end `ConsentGranted` event logged for compliance tracking.

Then upon getting the required consent the verifier(a certain bank where the customer is wanting to open an account) calls `verifyCredential()` with the original document hash.

```
function verifyCredential(bytes32 _credentialId, bytes32 _hash)
    public
    view
    returns (bool)
{
    Credential memory cred = credentials[_credentialId];
    return cred.credentialHash == _hash && !cred.isRevoked;
}
```

Smart contract then validates hash against stored credential. Then the function at the end returns verification status without exposing sensitive data.

8.4 Revocation Mechanism:

And finally if the original issuer of the credential sees that in the blockchain network one of its verified customer has been involved in any case that is violation in terms of the AML and CFT directives of BFIU and Bangladesh Bank, then that issuer can revoke the credentials of that user. It is to be noted that only the original issuer can revoke credentials. Revocation immediately invalidates credentials for all future verifications.

```
function revokeCredential(bytes32 _credentialId) public { 28134 gas
    Credential storage cred = credentials[_credentialId];
    require(msg.sender == cred.issuer, "Only issuer can revoke");
    cred.isRevoked = true;
    emit CredentialRevoked(_credentialId);
}
```

Then finally, **CredentialRevoked** event ensures audit trail continuity.

9. Privacy and Security Implementation:

Data Minimization: Our platform goes through privacy by design principles by storing only the required cryptographic hashes on chain. Actual KYC documents remain in encrypted off chain storage which is only accessible to authorized parties with proper consent.

Consent Management: The `grantConsent()` function creates a transparent, auditable consent mechanism where customers explicitly authorize each verification request.

Tamper proof Audit Trail: All credential operations generate blockchain events that create an immutable audit trail for regulatory compliance.

9.1 Setup and Execution:

We completed the code execution part from <https://remix.ethereum.org> . No local installation was required as remix operates completely from the browser. We chose the **Shanghai EVM** version since it is the closest to Bangladesh. Then from the Deploy transitions tab we deployed the contract and ran the code.

10. Scope of Work in CSE499B:

Upon the work we completed in CSE499A we will extend this to CSE499B. Our main goal in CSE499B will be developing a production ready eKYC platform which will attune to the needs for the Bangladeshi banking ecosystem. The primary deliverable will be a complete system demonstration ready for bank pitches, integrating regulatory compliance with Bangladesh Bank's AML-CFT guidelines and BFIU Circular 25 requirements.

We will go with the full stack development that will have a user-friendly web and mobile interfaces for all entities(customers, banks, regulators) with secure wallet integration and biometric authentication capabilities(bound to the first issuer as per our architecture). We will also have an enhanced contract integration where we will be implementing advanced features such as credential expiration tracking, regulatory reporting mechanisms and compliance with specific Bangladeshi documentation requirements such NID-integration, eTin validation and utility bill verification. We will be deep diving on how we can integrate the government databases to our system so that if a user is involved in any sort of violation or crime the blockchain network will be updated accordingly and banks can avoid that particular customer. If possible we will also try to onboard a real bank to do the pilot testing and validation. We have to check if everything is tamper proof and relevant performance valuation is good. And we will make sure that the developing APIs seamlessly integrate with existing banking systems. We will also keep a compliance reporting dashboards for BFIU requirements

and at the same time establish protocols for handling Enhanced Due Diligence (EDD) cases involving Politically Exposed Persons (PEPs).

11. Challenges:

While outlining our scope of work for CSE499B, it is also important to reflect on some of the practical challenges we faced during the initial development phase. One of the most significant hurdles was deciding on the most appropriate blockchain framework for our e-KYC system.

At first, we explored Hyperledger Indy, which was once considered a strong candidate for decentralized identity management due to its support for Self-Sovereign Identity (SSI) principles. Unfortunately, by 2025 Hyperledger Indy and its associated projects had lost active organizational and community support, making it unreliable for long-term adoption. This forced us to look for more sustainable alternatives.

After evaluating multiple options, we chose to proceed with the Ethereum blockchain, primarily due to its mature ecosystem, security, and wide-scale adoption in enterprise and financial applications. The use of Solidity, Ethereum's native smart contract language, provided us with robust tools to implement features like role-based access control, credential lifecycle management, and consent mechanisms. Furthermore, Ethereum's transition to Proof-of-Stake (PoS) aligned well with our vision of a collaborative and energy-efficient banking ecosystem, where stakeholder banks could participate fairly in the network governance.

By overcoming this early challenge, we ensured that our system's foundation is not only technically feasible but also future-proof and scalable, giving us the confidence to extend our work into CSE499B and beyond.

12. Conclusion:

The current KYC practices in Bangladesh remain largely manual and repetitive across different banks, resulting in inefficiency, higher costs and risks of fraud. Although a few banks have adapted partial digitalization, the lack of a well-maintained, secure and standardized system still limits their effectiveness. Our proposed blockchain-based e-KYC platform addresses these challenges by leveraging Ethereum smart contracts to ensure security, transparency and privacy of customers' verification data. By storing only cryptographic proofs on-chain and keeping sensitive documents off-chain, the system guarantees compliance with privacy requirements while maintaining tamper-proof audit trails for regulators. This approach significantly reduces customer onboarding time, prevents fraudulent activities through immutable verification records, and enhances customers privacy and satisfaction. Besides, it relieves banks from repetitive KYC checks and ensures consistency with existing regulatory frameworks under Bangladesh Bank and BFIU. While our implementation demonstrates the feasibility of such a system, future work will need to focus on wider adoption among banks, integration with national identity databases and stronger off-chain storage mechanisms to fully realize its potential. In conclusion, the smart contract-based e-KYC platform

represents a transparent, secure and efficient alternative to the traditional KYC processes in Bangladesh, with the potential to foster greater financial inclusion, reduce operational costs and build trust among all stakeholders in the banking ecosystem.

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